개인 보고서

한경식

실험-11. Input Lag 측정

디스플레이 종류	PDP		
제조사	삼성전자(주)	Model명	SPD-42C7HD
화면의 종횡비	16:09	화면의 크기	42 inch
Pixel Size	x	최대해상도	1204 x 768
콘트라스트비	10,000:1	휘도	1300 cd/m^2
디스플레이 종류	LCD		
제조사	하이얼 전자	Model명	HLT-42NHW25B
화면의 종횡비	16:09	화면의 크기	42 inch
Pixel Size	x	최대해상도	1920 x 1080
콘트라스트비	1,200:1	휘도	500 cd/m^2
디스플레이 종류	PDP		
휘도	1300 cd/m^2	컨트라스트	x
색온도	x	색감	표준
선명도	0		
디스플레이 종류	LCD		
휘도	500 cd/m^2	컨트라스트	57
색온도	normal	색감	표준
선명도	50		

Stop Watch

	CRT	PDP	LCD	CRT-PDP	CRT-LCD	LCD-PDP
1	00:01:45:321	00:01:45:271	00:01:45:281	50ms	40ms	10ms
2	00:01:45:951	00:01:45:901	00:01:45:921	50ms	30ms	20ms
3	00:01:46:051	00:01:46:001	00:01:46:021	50ms	30ms	20ms
4	00:01:46:151	00:01:46:101	00:01:46:121	50ms	30ms	20ms
5	00:01:46:251	00:01:46:201	00:01:46:221	50ms	30ms	20ms
6	00:01:46:352	00:01:46:302	12:01:46:322	50ms	30ms	20ms
7	00:01:46:552	00:01:46:502	00:01:46:522	50ms	30ms	20ms
8	00:01:46:582	00:01:46:532	00:01:46:552	50ms	30ms	20ms
9	00:01:46:722	00:01:46:672	00:01:46:682	0ms	30ms	10ms
10	00:01:46:782	00:01:46:732	00:01:46:752	50ms	30ms	20ms
11	00:01:46:882	00:01:46:832	00:01:46:852	50ms	30ms	20ms
12	00:01:46:982	00:01:46:932	00:01:46:952	50ms	30ms	20ms
13	00:01:47:082	00:01:47:032	00:01:47:052	50ms	30ms	20ms
14	00:01:47:152	00:01:47:102	00:01:47:122	50ms	30ms	20ms
15	00:01:47:202	00:01:47:152	00:01:47:172	50ms	30ms	20ms
16	00:01:47:272	00:01:47:222	00:01:47:232	50ms	30ms	10ms
17	00:01:47:302	00:01:47:232	00:01:47:272	70ms	30ms	30ms
18	00:01:47:322	00:01:47:272	00:01:47:282	50ms	40ms	10ms
19	00:01:47:332	00:01:47:282	00:01:47:302	50ms	30ms	20ms
20	00:01:47:352	00:01:47:302	00:01:47:322	50ms	30ms	20ms
21	00:01:47:382	00:01:47:332	00:01:47:352	50ms	30ms	20ms
22	00:01:47:402	00:01:47:352	00:01:47:372	50ms	30ms	20ms
23	00:01:47:422	00:01:47:372	00:01:47:382	50ms	40ms	10ms
24	00:01:47:472	00:01:47:402	00:01:47:432	70ms	40ms	30ms
25	00:01:47:552	00:01:47:502	00:01:47:522	50ms	30ms	20ms
26	00:01:47:572	00:01:47:522	00:01:47:532	50ms	40ms	10ms
27	00:01:47:582	00:01:47:522	12:01:47:552	60ms	30ms	0ms
28	00:01:47:602	00:01:47:532	00:01:47:572	70ms	30ms	40ms
29	00:01:47:622	00:01:47:572	00:01:47:582	50ms	40ms	10ms
30	00:01:47:632	00:01:47:582	00:01:47:602	50ms	30ms	20ms

SMTT

	CRT	PDP	LCD	CRT-PDP	CRT-LCD	LCD-PDP
1	27.646	27.600	27.615	0.046	0.031	0.015
2	28.287	28.226	28.25	0.061	0.037	0.061
3	28.453	28.325	28.351	0.128	0.102	0.026
4	28.478	28.427	28.451	0.051	0.027	0.024
5	28.611	28.527	28.552	0.084	0.059	0.025
6	28.833	28.711	28.786	0.122	0.047	0.075
7	28.946	28.811	28.853	0.135	0.065	0.042
8	29.019	28.91	28.954	0.109	0.065	0.044
9	29.142	29.017	29.046	0.125	0.096	0.029
10	29.442	29.379	29.423	0.063	0.019	0.044
11	29.643	29.579	29.623	0.064	0.02	0.044
12	29.755	29.679	29.713	0.076	0.042	0.034
13	29.907	29.782	29.824	0.125	0.083	0.042
14	30.044	30.014	30.038	0.03	0.006	0.024
15	30.12	30.083	30.096	0.037	0.024	0.013
16	30.173	30.131	30.152	0.042	0.021	0.021
17	30.203	30.165	30.178	0.038	0.025	0.013
18	30.231	30.181	30.209	0.05	0.022	0.028
19	30.241	30.198	30.232	0.043	0.009	0.034
20	30.251	30.215	30.239	0.036	0.012	0.024
21	30.267	30.232	30.258	0.035	0.009	0.026
22	30.283	30.25	30.265	0.033	0.018	0.015
23	30.296	30.265	30.282	0.031	0.002	0.017
24	30.306	30.265	30.294	0.041	0.012	0.029
25	30.318	30.282	30.306	0.036	0.012	0.024
26	30.341	30.318	30.329	0.023	0.012	0.011
27	30.331	30.3	30.329	0.031	0.002	0.029
28	30.362	30.316	30.344	0.046	0.018	0.028
29	30.37	30.332	30.356	0.038	0.014	0.024
30	30.407	30.367	30.394	0.04	0.013	0.027

고찰 및 검토

1) 결과값을 확인해 보면 CRT가 가장 빠르고 그 다음은 LCD, PDP순으로 느려지는 것을 볼 수 있다

2) Stop Watch의 평균 지연시간은 CRT-PDP:50.667, CRT-LCD:32, LCD-PDP:18.333 이다.

3) SMTT의 평균 지연시간은 CRT-PDP:0.060633, CRT-LCD:0.0308, LCD-PDP:0.0297 이다.

4) CRT는 간소화된 회로를 가졌기 때문에 이보다 복잡한 회로를 가진 LCD와 PDP가 지연시간이 생기는 것이다.

Q & A